# Lab 2 : Database System Design

Data Management Course
UM6P College of Computing
Professor: Karima Echihabi Program: Computer Engineering
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**Team Information**
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Repository Link: github.com/NJ-adam/data-Ahmad-Adam-Souyhayl-Ilyas.git

## 1 Introduction

The second lab consists of implementing the database schema using SQL. The script creates the `Lab2` database and defines the main entities and relationships needed to manage people, laboratories, research projects, budgets, equipment, certifications, reservations, maintenance, calibration, consumables and suppliers. This deliverable focuses on the relational implementation of the database and on enforcing the main structural constraints through primary keys, foreign keys and selected check constraints.

## 2 Requirements

The SQL implementation covers the following main requirements:

- Create the `Lab2` database and its relational tables.
- Represent people and their specializations through related tables such as `Student`, `Employee`, `Academic`, `Faculty`, `Non_Academic`, `Administrative` and `Technical`.
- Manage laboratories, research projects, participation and budgets.
- Represent equipment models, equipment units, certifications and reservations.
- Store maintenance and calibration records associated with equipment.
- Manage consumables, suppliers, laboratory stocks and consumption.

## 3 Team Work Distribution

The work was divided between the four members according to the main parts of the database:

| Member | Main work |
|---|---|
| **Adam Naji** | Research Project, Calibration records and Maintenance Operation |
| **Ahmad Khriss** | Laboratory and Equipment |
| **Souhayl Moujahid** | People, Reservations and Budgets |
| **Ilyas Mouhri** | Certifications and Consumables |

## 4 Methodology

We implemented the design as a relational schema where each main concept is represented by a table with a primary key. Foreign keys were used to connect dependent entities and relationships. For example, the specialization tables reference `people(pid)`, while `equipmentUnit` references both an equipment model and a laboratory.

Many-to-many relationships were represented using associative tables with composite primary keys, such as `participates`, `fundsprj`, `FundsLab`, `Reserve`, `requires` and `holds`. The same approach is used for laboratory stocks and supplier relations. Some allowed values are restricted directly in SQL using `CHECK` constraints, notably for project, equipment, reservation, maintenance and calibration statuses.

## 5 Implementation & Results

The implementation contains the complete set of table definitions and their relationships. The people hierarchy is represented through foreign-key chains, from `people` to `Employee` or `student`, and then to more specific employee categories. Laboratories are linked to faculty members, while projects can have participants and can be funded by budget lines.

For equipment, `equipmentModel` is separated from `equipmentUnit`. Each unit is associated with a laboratory and a model. Reservations are connected to equipment through `Reserve`. Maintenance and calibration records are linked directly to equipment units, with composite keys used to identify records.

The last part of the schema manages consumables. `stocks` connects consumables to laboratories and identifies the technical person responsible for the stock. `supplies` connects consumables, laboratories and suppliers, while `consumes` records the quantities used for reservations.

The script therefore implements the database structure and integrity constraints. Since the submitted SQL contains table definitions but no `INSERT`, `SELECT` or test queries, no populated-data results are reported here.

## 6 Discussion

One important point in the implementation was maintaining consistency between the different parts of the schema through foreign keys. Several tables also require composite primary keys because a record is identified by a combination of entities, for example a participant and a project or a reservation and an equipment unit.

The schema also shows the need to distinguish between general entities and their more specific categories. This is visible in the chain of foreign keys used for employees and in the separation between equipment models and physical equipment units.

## 7 Conclusion

The deliverable results in a relational SQL schema covering the main components of the database. Primary keys, foreign keys and selected domain checks were added to enforce structural and value constraints. The implementation provides the database foundation for storing and relating people, research activities, laboratories, equipment, reservations, maintenance, certifications and consumables.